Prob. 1: *Consider the circuit shown in Fig. 1,*

1. The load impedance (Z_L), in Ω, that absorbs the maximum average power is:

2. The open circuit voltage between *a-b* terminals (V_{th}), when the load impedance is disconnected, is:

3. The short circuit current (I_{sc}), when *a-b* terminals are shorted is:

4. The maximum average power transferred to the load impedance (Z_L) is:

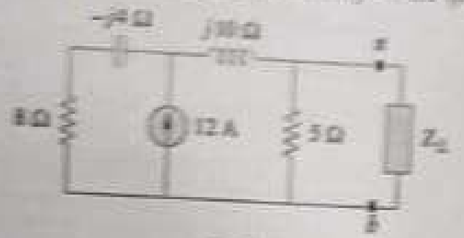

Fig.3

([illegible] marks)

Prob. 2: *Consider the circuit shown in Fig. 2 and using mesh analysis method,*

1. Write the two mesh equations, do not simplify:

2. The current I_1 is:

3. The current I_2 is:

4. The capacitor current, I_C, is:

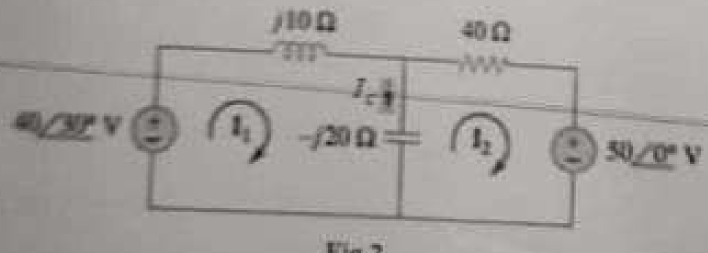

Fig 2

(8 marks)

Prob 3.: *Two loads are connected in parallel across a voltage source of $100\angle 0^{\circ} V_{rms}$, where load 1 absorbs 500VAR at 0.6 lagging p.f. while load 2 absorbs 400VA at 0.75 leading p.f*

1. The impedance of load 1 (Z_1), in Ω, is:

2. The current of load 2 (I_2), in Amps, is:

3. The complex power (S) supplied by the source is:

4. The power factor of the equivalent load is:

5. The capacitance needed to raise the pf to unity is:

6. The current supplied by the source, after the power factor is corrected, is:

(12 marks)

Prob. 4: *Consider the circuit shown in Fig. 3,*

1. The equivalent impedance seen by the source, Z_{in}, is:

2. The current supplied by the source, I_1, is:

3. The load current, I_2, is:

4. The voltage across the primary winding, V_1, is:

5. The average power absorbed by the 2Ω resistor is:

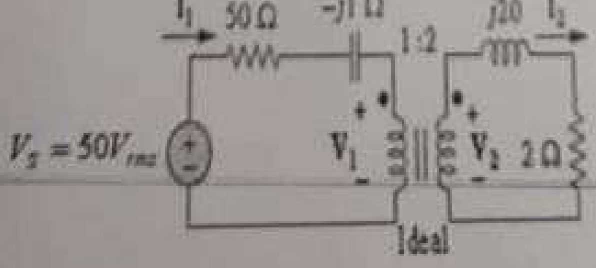

Fig.3

(10 marks)

Prob. 5: *Fig. 4 shows a balanced, positive sequence, Y-connected voltage source with $V_{an} = 240\angle 0^{\circ} V_{rms}$ and internal impedance of $Z_s = (0.4 + j0.3)\Omega$ per phase, connected to a Y- connected balanced load with an impedance of $Z_L = (24 + j19)\Omega$ per phase. If the line joining the source and the load has an impedance of $Z_l = (0.6 + j0.7)\Omega$ per phase,*

1. The line current I_{aA} is:

2. The load phase voltage V_{AN} is:

3. The load line voltage V_{AB} is:

4. The source line voltage V_{ab} is:

5. The complex power supplied by phase *a* of the source is:

6. The total average power absorbed by the load is:

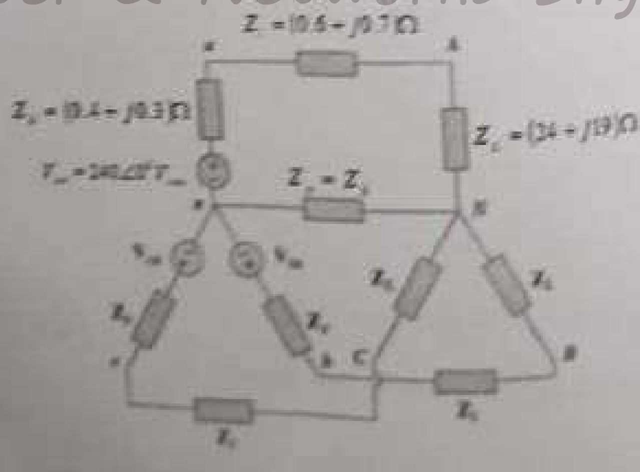

Fig. 4

(12 marks)

Good Luck